# Statistical Models, Supplementary Material

# Markov chains

## Markov chains in discrete time with finite or countable state space

In the introductory probability course ("ISS"), you learned about Markov chains $X_1, X_2, \ldots$ with a finite or countable state space, that is, the $X_i$ take only values from a finite or countable set $S$ of outcomes. Markov chains have the ***Markov property*** that says that the conditional probability of $X_{n+1}$ taking a value $x \in S$, given that $X_n = x_n$, does not depend on the previous values of $X_{n-1}, X_{n-2}, \ldots$, see for example Pishro-Nik, Section 11.2:

$$\mathrm{P}(X_{n+1} = x \mid X_n = x_n, X_{n-1} = x_{n-1}, \ldots) = \mathrm{P}(X_{n+1} = x \mid X_n = x_n)$$

A Markov chain with finite or countable state space is called ***homogeneous***, if its ***transition probabilities*** do not depend on the index $n$:

$$p_{ij} = \mathrm{Prob}(X_{n+1} = j \mid X_n = i).$$

To stretch that the homogeneity refers to shifting the index $n$, which is seen as "time", one also calls such Markov chains ***time-homogeneous***.

Sometimes, the starting value of the chain is denoted by $X_0$, in particular, when one wants to emphasize that this value is arbitrarily chosen by the user.

### Stationary and limiting distribution of a homogeneous Markov chain with finite state space

The distribution of the chain at "time" $n$, that is, the distribution of $X_n$, depends on the distribution of $X_{n-1}$ and thus indirectly on the distribution of $X_{n-2}$ etc. If it holds for a distribution $\pi$, that

$$X_n \sim \pi \implies X_{n+1} \sim \pi,$$

then we call that distribution a ***stationary distribution*** of the Markov chain.

For a Markov chain with finite state space that is irreducible, one can show that the stationary distribution is the ***limiting distribution***, that is, no matter what the value or distribution of $X_1$ is, the distribution of the chain converges to $\pi$:

$$\mathrm{P}(X_n \in A) \xrightarrow[n\to\infty]{} \pi(A) \quad \text{for all } A \subseteq S$$

that is,

$$X_n \xrightarrow[n\to\infty]{D} X, \quad \text{where } X \sim \pi.$$

see Pishro-Nik, Section 11.2.6

The concept of Markov chains as discussed above can be generalized in two ways: to continuous time chains, where we index the $X$es with a continuous variable $t \in \mathbb{R}$, or to continuous state space, such as $S \subset \mathbb{R}$ (or both).

## Markov chains with continuous state space

In the context of Markov chain Monte Carlo methods, we most often need ***continuous state spaces***, but stick to discrete time, that is indexing with integers.

For many random variables on $\mathbb{R}$, the probability of taking a given value $x$ is zero. E.g., when $X \sim N(0,1)$, we can safely say the $\text{Prob}(X = 0.2) = 0$. But for the same random variable, the probability of falling into the interval $[0, 0.2]$ is larger than zero. Instead of looking at the probability that $X_{n+1} = x$, we look at the probability $X_{n+1} \in A$ to define the Markov property for chains with continuous state space:

$$\text{P}(X_{n+1} \in A \mid X_n \in A_n, X_{n-1} \in A_{n-1}, \dots) = \text{P}(X_{n+1} \in A \mid X_n \in A_n),$$

where $A, A_n, A_{n-1}, \dots \subseteq S$ are any sets that make sense in this context (for those who have had a course on measure theory, they are "measureable sets").

### Transition (kernel) densities of homogeneous Markov chains

Transition probabilities as for Markov chains with countable state space are now replaced by ***transition kernel densities***. A kernel density $k$ is a nonnegative function on $S \times S$, and for each $x \in S$, $k(., x)$ is a probability density on $S$ that depends on $x$. In a time-homogeneous Markov chain, the transition kernel $k$ is the same for all times $n$. Then

$$\text{Prob}(X_{n+1} \in A \mid X_n \in B) = \int_B \int_A k(y, x) \text{d}y \, P_n(\text{d}x),$$

and $k(., x)$ is the probability density of $X_{n+1}$ when $X_n = x$, for any $n$. The inner integral, $\int_A k(y, x)\text{d}y$, can be interpreted as the conditional probability that $X_{n+1} \in A$ given that $X_n = x$. That integral can only take values between 0 and 1.

### Stationary and limiting distribution

The definition of stationary distribution carries right over from the case of Markov chains with finite or countable state space. Now, let $\pi$ be the stationary distribution of a Markov chain with transition density $k$, and assume that $X_n \sim \pi$. Then for all $y \in S$,

$$\pi(y) = \int_S k(y, x)\pi(x)\text{d}x.$$

This can be proven quite quickly using tools from measure theory, but more importantly, it has an intuitive explanation

For Markov chains with general state space (including continuous state space) $S$, we have a theorem that ensures that

> A homogeneous Markov chain $X_1, X_2, ...$, that has stationary distribution $\pi$, and is irreducible, aperiodic and Harris recurrent, always has $\pi$ as its limiting distribution. That is, the chain converges in distribution to $\pi$ irrespective of the starting value $X_0 = x_0$,
>
> $$\mathrm{P}(X_n \in A \mid X_0 = x_0) \xrightarrow[n\to\infty]{} \pi(A), \quad \text{for all } x_0 \in S.$$

The technical conditions of this theorem are usually fulfilled in MCMC applications. We will later deal with methods to check if this convergence holds, and to judge how fast the convergence is, see Reich & Ghosh, Section 3.4.